Final Project Report — Part 1: Overview

SmartScan Image Processing Evaluation Document

Introduction: The Smart Document Scanner project presents a real-time computer vision workflow capable of converting unconstrained mobile phone captures into high-resolution scanner quality outputs.

Motivation: Handheld captures suffer from perspective keystone distortion, non-uniform ambient illumination, and motion blur. Traditional flatbed scanners eliminate these physical anomalies through controlled optical geometry.

Objectives: 1. Accurate quadrangular paper contour localization. 2. Inverse projective homography mapping. 3. Morphological illumination compensation. 4. High-accuracy OCR transcription.

Final Project Report — Part 2: Methodology

SmartScan Image Processing Evaluation Document

Mathematical Foundation: An arbitrary planar surface photographed under projective perspective transforms planar coordinates via a 3x3 homography matrix H with 8 degrees of freedom.

Edge Processing: The Canny edge detector applies Gaussian smoothing followed by Sobel gradient computations, non-maximum suppression, and hysteresis thresholding.

Illumination Correction: Background estimation employs morphological dilation using a circular structuring element followed by median blur filtering.

Final Project Report — Part 3: Evaluation

SmartScan Image Processing Evaluation Document

Experimental Results: The pipeline was evaluated against synthetic benchmark scenarios representing various environmental distortions.

Optical Character Recognition: Evaluated using Tesseract OCR v5. Confidence scores and word recognition rates demonstrate superior fidelity on shadow-removed and CLAHE-enhanced outputs.

Conclusion: The system delivers competitive performance relative to commercial mobile scanning solutions.